

BIOCHEMISTRY (LAS)

Overview

The Roy J. Carver Department of Biochemistry, Biophysics and Molecular Biology (<https://www.bbmb.iastate.edu>) offers majors in biochemistry in the College of Liberal Arts and Sciences and in the College of Agriculture and Life Sciences. Biochemists seek to understand life processes in terms of chemical and physical principles. Students in the biochemistry majors develop foundational analytical skills while exploring frontiers in biotechnology and medicine. Graduates in biochemistry will have a rigorous background in chemistry, biology, and physics. Biochemistry coursework focuses on the development of problem-solving skills, critical thinking, communication, and research design. The Biochemistry degree includes focused specializations towards professional goals, including three prescribed options: **Research & Biotechnology**, **Pre-Medicine**, and **Biophysics**.

Graduates with a Biochemistry B.S. degree will be prepared for postgraduate studies in the chemical or biological sciences, medical and health professional training, or immediate laboratory research in biochemistry, biotechnology, or pharmacy. Graduates are also equipped to pursue careers in teaching, technical writing, science communication and policy, intellectual property law, and biotechnology entrepreneurship. The biochemistry major is accredited by the American Society for Biochemistry and Molecular Biology (ASBMB). As such our learning objectives align with ASBMB core concepts.

Student Learning Outcomes

Upon graduation, students should be able to:

- Explain and provide examples to illustrate the biochemical principles underlying the following:
 - a. How energy is required and transformed in biological systems.
 - b. How macromolecular structure determines function and regulation.
 - c. How information storage and flow are dynamic and interactive.
 - d. How biochemistry and biology are driven by evolution and homeostasis
- Apply and justify appropriate techniques to characterize and quantify biomolecules in biological systems.
- Execute effective multimodal communication of the research process and results.
- Implement and justify best practices for laboratory safety and research ethics.

Biochemistry Major in the College of Liberal Arts and Sciences

The Bachelor of Science (B.S.) degree in Biochemistry requires the Biochemistry Core and one of the following three prescribed options. The options represent focused specializations for professional goals, including *Research & Biotechnology*, *Pre-Medicine*, and *Biophysics*. As majors in the College of Liberal Arts and Sciences, Biochemistry students must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to those stated below. Focus option-approved science course lists can be found on the Biochemistry website (<https://www.bbmb.iastate.edu/approved-course-lists/>).

Biochemistry Core

The major in biochemistry requires completion of the Biochemistry Core and one of three focus options. Each defined option includes specific supporting coursework in Biology; Chemistry; Mathematics and/or Statistics.

BBMB 1010	Introduction to Biochemistry	2
BBMB 1020	Introduction to Biochemistry Laboratory	1
BBMB 2010	Chemical Principles in Biological Systems	2
BBMB 3110	Writing Scientific Reports in Biochemistry ¹	1
BBMB 3120	Experimental Research Skills in Biochemistry	2
BBMB 4040	Biochemistry I	3-4
or BBMB 5040 & BBMB 5050	Amino Acids and Proteins and Bioenergetics and Metabolism	
BBMB 4050	Biochemistry II	3-4
or BBMB 5060 & BBMB 5070	Membrane Biochemistry and Biochemistry of Nucleic Acids	
BBMB 4100	Analysis of Scientific Literature	2
BBMB 4110	Techniques in Biochemical Research	4
BBMB 4610	Molecular Biophysics	2
BBMB 5610L	Laboratory in Molecular Biophysics	2-3
or CHEM 3220L	Laboratory in Physical Chemistry	
PHYS 2310 & 2310L	Introduction to Classical Physics I and Introduction to Classical Physics I Laboratory	5
PHYS 2320 & 2320L	Introduction to Classical Physics II and Introduction to Classical Physics II Laboratory	5
BBMB 4990	Undergraduate Research (highly encouraged elective)	arr †

BBMB 4970	Biochemistry Program Outcomes Assessment	R
Total Credits	34-37	†

† Arranged with instructor.

¹ BBMB 3110 fulfills the ISU upper-level Communication Proficiency requirement.

Biochemistry Program of Study: Research & Biotechnology

Students interested in careers in biotechnology research and/or graduate studies in biochemistry, chemistry, or bioscience fields may opt for the Research & Biotechnology Option.

In addition to the Biochemistry Core (above):

CHEM 2010	Advanced General Chemistry	5-7
or CHEM 1770 & CHEM 1780	General Chemistry I and General Chemistry II	
CHEM 2010L	Laboratory in Advanced General Chemistry	1
or CHEM 1770L or CHEM 1770LL	Laboratory in General Chemistry I Laboratory in General Chemistry I	
CHEM 3250	Chemical Thermodynamics	3
CHEM 3310	Organic Chemistry I	3
CHEM 3320	Organic Chemistry II	3
CHEM 3330L	Laboratory in Organic Chemistry I (for Chemistry and Biochemistry Majors)	1-2
or CHEM 3310L	Laboratory in Organic Chemistry I	
CHEM 3340L	Laboratory in Organic Chemistry II (for Chemistry and Biochemistry Majors)	1-2
or CHEM 3320L	Laboratory in Organic Chemistry II	
MATH 1650 & MATH 1660	Calculus I and Calculus II	8
MATH 2650	Calculus III	3-4
or MATH 2660 or MATH 2670	Elementary Differential Equations Elementary Differential Equations and Laplace Transforms	
or STAT 2010 or STAT 3005	Introduction to Statistical Concepts and Methods Engineering Statistics	
BIOL 2110	Principles of Biology I	3
BIOL 2120	Principles of Biology II	3
BIOL 3130 & 3130L	Principles of Genetics and Genetics Laboratory	4
BIOL 3140	Principles of Molecular Cell Biology	3

Molecular Sciences from approved list	3
Total Credits	44-49

Biochemistry Program of Study: Pre-Medicine

Students interested in qualifying for medical school training for careers as a Doctor of Medicine (MD) or Doctor of Osteopathic Medicine (DO) may opt for the Pre-Medicine Option. Allied health professions (e.g., Physician Assistant, Dentistry, and Ophthalmology) may also consider the Pre-Medicine Option.

In addition to the Biochemistry Core (above):

CHEM 1770 & 1770L & CHEM 1780 & CHEM 1780L	General Chemistry I and Laboratory in General Chemistry I and General Chemistry II and Laboratory in College Chemistry II	6-9
or CHEM 2010 & 2010L	Advanced General Chemistry and Laboratory in Advanced General Chemistry	
CHEM 3250	Chemical Thermodynamics	3
CHEM 3310 & 3310L	Organic Chemistry I and Laboratory in Organic Chemistry I	4
CHEM 3320 & 3320L	Organic Chemistry II and Laboratory in Organic Chemistry II	4
MATH 1650 & MATH 1660	Calculus I and Calculus II	8
STAT 2010 or STAT 3005	Introduction to Statistical Concepts and Methods Engineering Statistics	3-4
PSYCH 1010 or PSYCH 2300	Introduction to Psychology Developmental Psychology	3
SOC 1340	Introduction to Sociology	3
BIOL 2110	Principles of Biology I	3
BIOL 2110L	Principles of Biology Laboratory I	1
BIOL 2120	Principles of Biology II	3
BIOL 2120L	Principles of Biology Laboratory II	1
BIOL 3130	Principles of Genetics	3
BIOL 3140	Principles of Molecular Cell Biology	3
Health & Biosciences from approved list		3
Total Credits		51-55

Biochemistry Program of Study: Biophysics

Students interested in research careers focused on the biophysical basis of life processes may opt for the Biophysics Option.

In addition to the Biochemistry Core (above):

CHEM 2010	Advanced General Chemistry	5-7
or CHEM 1770	General Chemistry I	
& CHEM 1780	and General Chemistry II	
CHEM 2010L	Laboratory in Advanced General Chemistry	1
or CHEM 1770N	Laboratory in General Chemistry I	
or CHEM 1770L	Laboratory in General Chemistry II	
CHEM 3240	Introductory Quantum Mechanics	3
CHEM 3250	Chemical Thermodynamics	3
CHEM 3310	Organic Chemistry I	3
CHEM 3320	Organic Chemistry II	3
MATH 1650	Calculus I	8
& MATH 1660	and Calculus II	
MATH 2650	Calculus III	4
MATH 2660	Elementary Differential Equations	3
MATH 3170	Theory of Linear Algebra	4
COMS 2070	Fundamentals of Computer Programming	3
BIOL 2110	Principles of Biology I	3
BIOL 2120	Principles of Biology II	3
Physical Sciences from approved list		3
Total Credits		49-51

As majors in the College of Liberal Arts and Sciences, Biochemistry students must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to the requirements for the major.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level. You must also complete the LAS world language requirement and LAS career proficiency requirement.

Communication Proficiency requirement: According to the university-wide Communication Proficiency Grade Requirement (<https://catalog.iastate.edu/academics/#communicationproficiencytext>), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (<http://www.registrar.iastate.edu/courses/div-ip-guide.html>) for a list of approved courses. Discuss with your advisor how the two courses that you select can be applied to your graduation plan.

Four Year Plans

Biochemistry B.S. Program of Study, College of Liberal Arts and Sciences

Research & Biotechnology Option

Sample Four-Year Plan

First Year

Fall	Credits	Spring	Credits
BBMB 1010	2	BBMB 1020	1
BBMB 1100	1	BBMB 1110	1
MATH 1650	4	MATH 1660	4
CHEM 1770	4	CHEM 1780	3
CHEM 1770N	1	BIOL 2110	3
LIB 1600	1	General Education Elective	3
ENGL 1500	3		
		16	15

Second Year

Fall	Credits	Spring	Credits
MATH 2650, 2660, 2670, STAT 2010, or STAT 3005	3-4	BBMB 2010	2
CHEM 3310	3	CHEM 3320	3
CHEM 3310L	1	CHEM 3320L	1
PHYS 2310	4	PHYS 2320	4
PHYS 2310L	1	PHYS 2320L	1
BIOL 2120	3	ENGL 2500	3
		BBMB 4990	1
		LAS 2030	1
		15-16	16

Third Year

Fall	Credits	Spring	Credits
BBMB 4040	3	BBMB 4050	3
BBMB 3110*	1	BIOL 3140	3
BBMB 3120	2	BBMB 4100	2
BIOL 3130	3	BBMB 4990	1
BIOL 3130L	1	International Perspectives Elective	3
BBMB 4990	1	General Education Elective	3
U.S. Cultures and Communities Elective	3		
General Education Elective	3		
		17	15

Fourth Year

Fall	Credits	Spring	Credits
BBMB 4110	4	CHEM 3250	3

BBMB 4990	2 CHEM 3220L or BBMB 4610 <i>and</i> BBMB 5610L	3-4	BBMB 3120	2 BBMB 4100	2
Upper-level Mol. Sciences Elective	3 BBMB 4990	3	BIOL 3130	3 BBMB 4990	1
General Education Elective	3 General Education Elective	3	STAT 2010	4 SOC 1340	3
General Education Elective	3 General Education Elective	3	PHIL 2350 ¹	3 International Perspectives Elective	3
	BBMB 4970	Required		16	15

15

15-16

"General Education Electives" include credits in LAS-approved courses in *Arts & Humanities* and *Social Sciences*

* BBMB 3110 fulfills the upper-level communication proficiency requirement.

Biochemistry B.S. Program of Study, College of Liberal Arts and Sciences *Pre-Medicine Option* Sample Four-Year Plan

First Year

Fall	Credits Spring	Credits
BBMB 1010	2 BBMB 1020	1
BBMB 1100	1 BBMB 1110	1
MATH 1650	4 MATH 1660	4
CHEM 1770	4 CHEM 1780	3
CHEM 1770N	1 CHEM 1780L	1
LIB 1600	1 BIOL 2110	3
ENGL 1500	3 BIOL 2110L	1
	16	14

Second Year

Fall	Credits Spring	Credits
CHEM 3310	3 BBMB 2010	2
CHEM 3310L	1 CHEM 3320	3
PHYS 2310	4 CHEM 3320L	1
PHYS 2310L	1 PHYS 2320	4
BIOL 2120	3 PHYS 2320L	1
BIOL 2120L	1 ENGL 2500	3
PSYCH 1010	3 LAS 2030	1
	16	15

Third Year

Fall	Credits Spring	Credits
BBMB 4040	3 BBMB 4050	3
BBMB 3110*	1 BIOL 3140	3

Fourth Year

Fall	Credits Spring	Credits
BBMB 4110	4 CHEM 3250	3
BBMB 4990	2 CHEM 3220L or BBMB 4610 <i>and</i> BBMB 5610L	3-4
Upper-level Health or Mol. Biosci. Elective	3 BBMB 4990	3
General Education Elective	3 General Education Elective	3
General Education Elective	3 General Education Elective	3
	BBMB 4970	Required
	15	15-16

"General Education Electives" include credits in LAS-approved courses in *Arts & Humanities* and *Social Sciences*. MCAT-recommended SOC 1340 and PSYCH 1010 apply towards *Social Sciences* requirement.

* BBMB 3110 fulfills the upper-level communication proficiency requirement.

¹ Fulfills university U.S. Cultures and Communities requirement

Biochemistry B.S. Program of Study, College of Liberal Arts and Sciences *Biophysics Option* Sample Four-Year Plan

First Year

Fall	Credits Spring	Credits
BBMB 1010	2 BBMB 1020	1
BBMB 1100	1 BBMB 1110	1
MATH 1650	4 MATH 1660	4
CHEM 1770	4 CHEM 1780	3
CHEM 1770N	1 BIOL 2110	3
LIB 1600	1 General Education Elective	3
ENGL 1500	3	
	16	15

Second Year

Fall	Credits Spring	Credits
MATH 2650	4 BBMB 2010	2
CHEM 3310	3 CHEM 3320	3

PHYS 2310	4	PHYS 2320	4
PHYS 2310L	1	PHYS 2320L	1
BIOL 2120	3	ENGL 2500	3
		MATH 2660	3
		LAS 2030	1
15		17	

Third Year

Fall	Credits	Spring	Credits
BBMB 4040	3	BBMB 4050	3
BBMB 3110*	1	CHEM 3250	3
BBMB 3120	2	BBMB 4100	2
COMS 2070	3	MATH 3170	4
U.S. Cultures and Communities Elective	3	BBMB 4990	1
General Education Elective	3	General Education Elective	3
15		16	

Fourth Year

Fall	Credits	Spring	Credits
BBMB 4110	4	CHEM 3240	3
STAT 3005	3	CHEM 3220L or BBMB 4610 and BBMB 5610L	3-4
BBMB 4990	1	Upper-level Physical Sci. Elective	3
International Perspectives Elective	3	BBMB 4990	1
General Education Elective	3	General Education Elective	3
General Education Elective	3	General Education Elective	3
		BBMB 4970	Required
17		16-17	

"General Education Electives" include credits in LAS-approved courses in *Arts & Humanities* and *Social Sciences*.

* BBMB 3110 fulfills the upper-level communication proficiency requirement.

Minor

The Roy J. Carver Department of Biochemistry, Biophysics, and Molecular Biology offers a minor in Biochemistry, administered by the College of Liberal Arts and Sciences.

BBMB 4040	Biochemistry I	3
BBMB 4050	Biochemistry II	3
BBMB 3120	Experimental Research Skills in Biochemistry	2
One course from the following:		2-3

BBMB 4610	Molecular Biophysics (2 cr)	
BBMB 5610	Molecular Biophysics (2 cr)	
CHEM 3250	Chemical Thermodynamics (3 cr)	
3000+ level courses in BBMB or CHEM to 15 cr total		5-6
Total Credits		15-17

All minors require at least 15 credits. At least 9 credits of the minor must be in courses numbered 2000 or above, of which at least 6 credits number 3000 or above. At least 3 credits applied to the minor must be taken at Iowa State University. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement.

The Biochemistry Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000 and above, with a grade of C or higher.

Concurrent Programs

Concurrent Bachelor of Science (B.S.)/ Master of Science (M.S.) Degrees

The department offers a concurrent enrollment degree program in either Biochemistry or Biophysics that allows ISU undergraduate students to obtain both the B.S. and M.S. degrees in about five years. The program is open to undergraduate students in the College of Liberal Arts and Sciences and in the College of Agriculture and Life Sciences. The concurrent degrees can be useful to students entering various career tracks. For those considering careers as research specialists, entry positions with higher-level responsibilities, and a higher-level salary, are made possible with the M.S. degree. For those considering careers as research directors, which require advanced study, the M.S. degree provides an advantage for admission into Ph.D. programs at the most competitive and prestigious graduate schools. Similarly, the M.S. degree can be a competitive advantage for admission in to medical, dental, law, veterinary medicine, or other professional schools.

Application to the program is made near the end of the junior undergraduate (third) year. Concurrent B.S./M.S. degree students begin research for the M.S. thesis during the summer semester after their junior year and are eligible for research assistantships, which are renewable based on academic standing and satisfactory research performance. The M.S. thesis requires intensive experience in original, independent laboratory research under the close supervision of a faculty mentor. To apply, see the concurrent B.S./M.S. application instructions found on the department's Graduate Study web page.

Concurrent Bachelor of Science/Graduate Certificate

The Bachelor of Science /Graduate Certificate program is intended for exceptional undergraduate students majoring in Biochemistry. In this

program, the student completes all of the requirements for the B.S. degree and the graduate certificate in a four-year period by combining the requirements of the two programs. The student enters the Graduate College after he/she achieves junior status and develops a plan of coursework (graduate and undergraduate) subject to the approval of the Director of Certificate (DOC). Required graduate courses are BBMB 5040, 5050, 5060, 5070, 5610 and 5610L. The student must satisfy the requirements of the B.S. in Biochemistry (121 credits) and the Graduate Certificate in Biochemistry (12 credits). Six credits of graduate coursework can satisfy some requirements of the B.S. degree. To apply for the B.S./Graduate Certificate, submit the application form found on the Graduate College Forms web page.

Graduate Programs

Introduction

Biochemistry and Biophysics are the science and technology used to understand the mechanisms underlying biological processes at the molecular level, with an emphasis on the fundamental relationships among the chemical, physical, and biological sciences. The Roy J. Carver Department of Biochemistry, Biophysics, and Molecular Biology (BBMB) administers Doctor of Philosophy (Ph.D.), Master's (M.S.), and Graduate Certificate programs that lead to an advanced degree or certificate in these disciplines. The prerequisite to graduate study is a sound undergraduate background in biology, chemistry, mathematics, and physics.

BBMB offers Doctor of Philosophy and Master's degrees in Biochemistry and in Biophysics that are designed to train students to independently conceive and carry out original research. BBMB also offers two graduate certificate programs in Biochemistry that provide a mechanism for formal recognition of focused graduate study in a specialized area that is less comprehensive than that required for a master's degree. BBMB participates in the Interdepartmental majors of Bioinformatics and Computational Biology; Genetics and Genomics; Immunobiology; Molecular, Cellular, and Developmental Biology; Neuroscience; Plant Biology; and Toxicology. All graduate degree students in BBMB are required to teach as part of their training.

Master of Science (M.S.) Degree

The M.S. degree programs in Biochemistry and in Biophysics are useful for students who prefer to undertake research training without the longer-term commitment required for the Ph.D. degree. It is also useful for students interested more in the technical aspects of research rather than in careers as research directors. The program requires about 3 years on average to complete and the successful defense of an independent research dissertation is required. About half the time required to earn the degree is spent on advanced coursework and professional seminars, and the other half is devoted to research undertaken in the laboratory under the close supervision of a faculty mentor. Financial support is available.

To apply, applicants first submit the free BBMB online application found on the department website, which is used as a screening tool.

NOTE: Students interested in a research career are encouraged to consider the Ph.D. track. Students may enter the Biochemistry or Biophysics M.S. degree program as a direct admit to a faculty research group at any time during the year.

Doctor of Philosophy (Ph.D.) Degree

The Ph.D. programs in Biochemistry and in Biophysics are designed to train students in the ability to independently conceive and carry out original research in the general area of the chemistry or physics of the processes of life. The programs require about 5-6 years on average to complete and the successful defense of an independent research dissertation. The majority of the time required to earn the degree is spent doing research on the dissertation project in the laboratory under the close supervision of a faculty mentor. Considerable time also is devoted to advanced coursework and professional seminars. Financial support is available. To apply, applicants first submit the free BBMB online application found on the department website, which is used as a screening tool. Students may enter the Biochemistry or Biophysics Ph.D. degree programs either as a rotation student in the fall semester or as a direct admit to a faculty research group at any time during the year.

Graduate Minor in Biochemistry

Graduate students in other M.S. and/or Ph.D. programs at ISU can earn a graduate minor in Biochemistry by completing 12 credits of the following courses with a grade point average of 3.0 or above: at least 6 credits from BBMB 5040, BBMB 5050, BBMB 5060 and BBMB 5070 and at least 6 credits of other BBMB 5000- and 6000-level courses. A student wishing to declare a minor in Biochemistry should arrange for a member of the graduate faculty in Biochemistry to serve on the POS Committee and submit the required form found on the Graduate College Forms page.

Graduate Certificate Programs

The graduate certificate program is designed for students who wish to continue or expand their knowledge in Biochemistry at the graduate level without the time commitment or lab experience required for a formal Master's or Ph.D. program. A certificate program can be an attractive option for individuals who have a bachelor's degree and are interested in broadening their expertise, or who are working in the sciences or industry and want to continue their education at the graduate level. BBMB offers two graduate certificate programs in Biochemistry: a concurrent B.S. / Graduate Certificate and a Graduate Certificate. The standards of admission and the course standards to which a certificate student is held are equivalent to those expected of a master's student. Each graduate certificate requires at least 12 graduate credits, all of which are available either on campus or online. A graduate supervisor will be appointed to oversee the certification for each student.

If a person who completes a graduate certificate program decides to continue for a graduate degree in Biochemistry or Biophysics, program approval is required. Credits earned for the graduate certificate may be used to meet course requirements for the graduate degree program.

Graduate Certificate in Biochemistry

The graduate certificate in Biochemistry is designed for students who have a B.S. degree in Biochemistry or a related field and wish to advance their knowledge by taking additional coursework at the graduate level.

The graduate certificate courses may be taken either on-line or on campus. Candidates for a graduate certificate in Biochemistry are admitted in the Graduate College. A total of 12 credits is required that includes: BBMB 5040, BBMB 5050, BBMB 5060 and BBMB 5070, plus four additional credits of BBMB coursework at the 5000-level. The 12 credits earned in the graduate certificate program may be applied to meet the course requirements of a M.S. or Ph.D. program in Biochemistry at Iowa State University (ISU) if the student is accepted into one of these programs. To apply for the graduate certificate in Biochemistry, submit the ISU online application.